Polynomial Regression

Degree Selection and Regularisation — Roll Number: BT2024108

1. Objective

The objective is to build polynomial regression models for two personalised datasets and generate predictions for their corresponding test sets. Rather than selecting a regulariser in advance, three polynomial regression approaches were compared: ordinary Linear Regression, Ridge regression and Lasso regression. The final model for each dataset was selected using cross-validation.

2. Dataset and Methodology

Each dataset contains 1000 training samples and 1000 test samples with target column y. For var1, six input features (x1–x6) are used; for var2, three input features (x1–x3) are used. Polynomial degrees 1–10 were evaluated for var1 and 1–20 for var2.

Approach	Pipeline
Polynomial + Linear	PolynomialFeatures → StandardScaler → LinearRegression
Polynomial + Ridge	PolynomialFeatures → StandardScaler → Ridge
Polynomial + Lasso	PolynomialFeatures → StandardScaler → Lasso

For every degree, Linear Regression was evaluated directly. Ridge and Lasso used $\alpha \in \{0.01, 0.1, 1, 10\}$. Standardisation was kept inside the modelling pipeline, so each cross-validation training fold fitted its own scaler and no validation-fold information leaked into training. Five-fold cross-validation used shuffled folds with random_state = 42. The primary selection criterion was the lowest mean validation MSE; mean R^2 was retained as a supporting metric.

3. Model Selection Results

Dataset	Best method	Degree	α	CV MSE	CV R^2
var1	Lasso	5	0.01	0.321413	0.965634
var2	Ridge	11	1	0.238684	0.994651

For var1, Lasso at degree 5 achieved the lowest validation MSE (0.321413), outperforming the best ordinary polynomial model (degree 4, MSE 0.713839) and the best Ridge configuration in the evaluated grid (degree 5, α=10, MSE 0.512885). For var2, Ridge at degree 11 and α=1 achieved the lowest validation MSE (0.238684), improving on the best ordinary polynomial model (degree 8, MSE 0.295215).

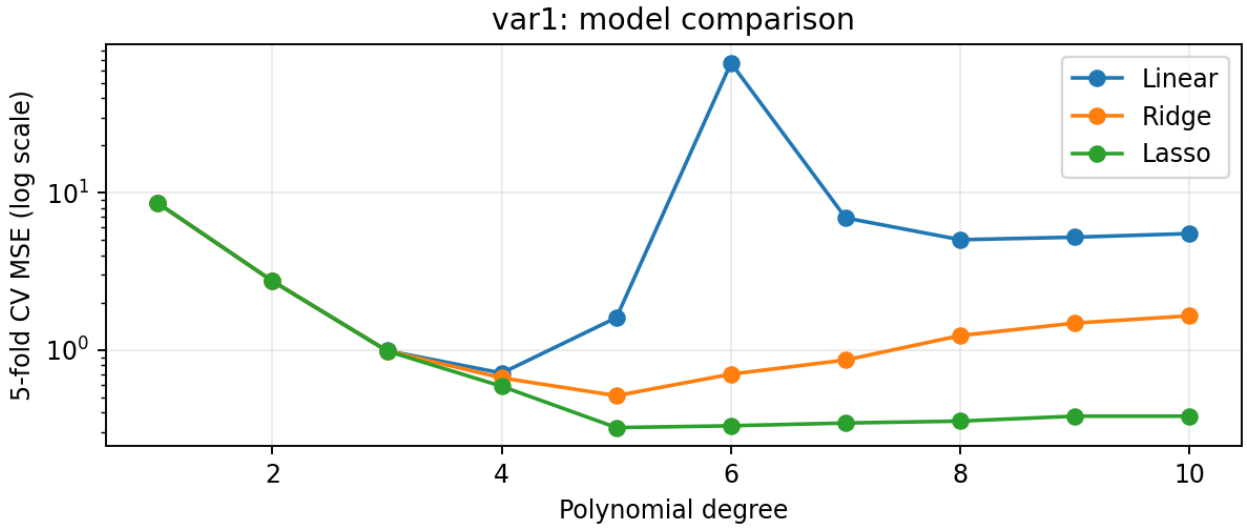


Figure 1. var1 cross-validation MSE by polynomial degree; regularised methods remain much more stable at higher degrees.

4. Phase 1 — var1

The var1 dataset contains six predictors, so the polynomial expansion grows quickly with degree. Ordinary polynomial regression performs well around degree 4 but becomes unstable at degree 6 and above. Ridge controls this instability but does not match Lasso in the selected grid. Lasso gives its best result at degree 5 with α=0.01 and retains a strong validation R^2 of 0.965634. The lower MSE indicates that the sparse regularisation is beneficial for this personalised dataset.

The model used for final var1 inference was therefore PolynomialFeatures(degree=5) + StandardScaler + Lasso(α=0.01), refitted on all 1000 training samples.

5. Phase 2 — var2

For var2, degrees 1–20 were evaluated. The validation error falls sharply as the polynomial degree increases and reaches its best overall value with Ridge at degree 11 and α=1. The ordinary polynomial baseline begins to deteriorate after its best region, while Ridge maintains controlled validation error at higher degrees. Lasso also remains comparatively stable but has a higher minimum MSE than Ridge.

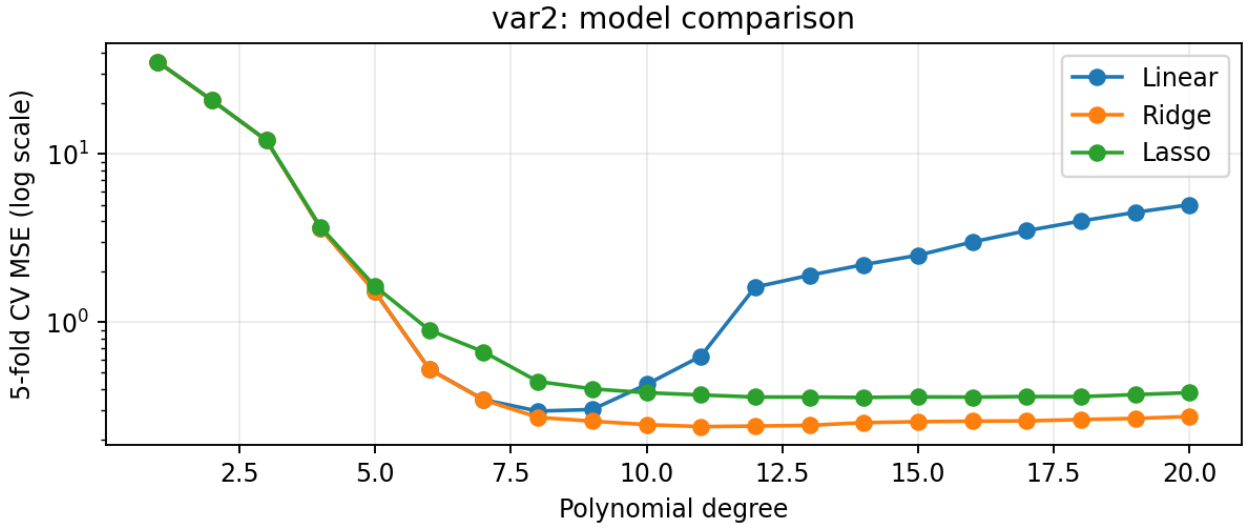


Figure 2. var2 cross-validation MSE by polynomial degree (log scale).

6. Final Models and Prediction Generation

Dataset	Final model	Degree	α	CV MSE	CV R^2
var1	Lasso	5	0.01	0.321413	0.965634
var2	Ridge	11	1	0.238684	0.994651

After selection, each winning pipeline was refitted using the complete corresponding training set. Predictions were generated for all 1000 test samples. The submission files contain exactly one column named y and no index:

- BT2024108_pred_var1.csv
- BT2024108_pred_var2.csv

7. Reproducibility and Repository

All training, cross-validation and inference code, personalised datasets, fitted models, results and this report are included in the GitHub repository:

https://github.com/car-ui/ML-Assignment

The main scripts are evaluate.py for model comparison, train.py for fitting the selected models on all training data, and predict.py for generating the required CSV files. The dependency list is provided in requirements.txt.

8. Conclusion

The comparison confirms that the best regularisation method depends on the personalised dataset. Lasso performs best for var1, while Ridge performs best for var2. Selecting among Linear, Ridge and Lasso using the same five-fold cross-validation protocol avoids assuming a particular regulariser beforehand. The final models were chosen by validation MSE, then retrained on all available training data before producing the test predictions.

Final submission files

1. BT2024108_pred_var1.csv
2. BT2024108_pred_var2.csv
3. BT2024108_Polynomial_Regression_Report.pdf

GitHub: https://github.com/car-ui/ML-Assignment